Quantum Gates: Fundamentals & Operations

Building Blocks of Quantum Circuits & Reversible Computation • Guide (0WZmklyHOks)

1. Introduction to Quantum Gates

In classical computing, logic gates (AND, OR, NOT) manipulate classical bits (0 or 1) to execute algorithm steps [cite: 1.1.1, 1.3.6]. In quantum computing, **quantum gates** are the fundamental operators that manipulate qubit state vectors to perform calculations [cite: 1.1.1, 1.3.6].

- **Linear Transformations:** Quantum gates act as linear operators on quantum states, represented mathematically by **unitary matrices** [cite: 1.3.4, 1.3.6].
- **Reversibility Principle:** Unlike many classical logic gates (such as AND or OR gates which destroy input information), all quantum gates are inherently **reversible** [cite: 1.1.1, 1.3.6]. Reversibility ensures that no quantum information is lost during computation [cite: 1.1.1, 1.3.6].

2. Fundamental Single-Qubit Gates

Single-qubit gates alter the state of an individual qubit by modifying its probability amplitudes or phase angle [cite: 1.1.1, 1.3.5].

Pauli-X Gate (Quantum NOT)

Flips the computational basis state [cite: 1.1.1, 1.3.6]:
$X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$.
Swaps the probability amplitudes α and β [cite: 1.1.1, 19].

Hadamard (H) Gate

Creates an equal superposition state [cite: 1.1.1, 1.3.6]:
$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$
Transforms deterministic basis states into superpositions [cite: 1.1.1, 1.3.6].

Pauli-Z Gate (Phase Flip)

Leaves $|0\rangle$ unchanged but flips the phase of $|1\rangle$ [cite: 1.1.1, 1.3.6]:
$Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$.
Alters quantum interference without changing outcome probabilities [cite: 1.1.1, 19].

Pauli-Y Gate

Combines a bit flip and phase flip [cite: 1.3.6]:
$Y|0\rangle = i|1\rangle$ and $Y|1\rangle = -i|0\rangle$.
Introduces complex phase factors into state vectors [cite: 1.3.6].

3. Multi-Qubit Gates & Entanglement Generation

Multi-qubit gates allow qubits to interact with each other, forming multi-qubit correlations essential for quantum speedups [cite: 1.1.1, 1.3.5, 1.3.6].

DIAGRAM: CNOT GATE & ENTANGLEMENT CREATION

Controlled-NOT (CNOT) Logic:
- **Control Qubit = |0⟩:** Target qubit remains unchanged [cite: 1.3.5]. ($|00\rangle \rightarrow |00\rangle$, $|01\rangle \rightarrow |01\rangle$)
- **Control Qubit = |1⟩:** Target qubit is flipped [cite: 1.3.5]. ($|10\rangle \rightarrow |11\rangle$, $|11\rangle \rightarrow |10\rangle$)

Creating a Bell State (Entangled Pair):
Input state $|00\rangle$ → Apply **H Gate** to Qubit 1 → $(|0\rangle + |1\rangle)|0\rangle / \sqrt{2}$ → Apply **CNOT** → $(|00\rangle + |11\rangle) / \sqrt{2}$ [cite: 1.1.1, 1.3.5, 1.3.6].

4. Quantum Circuits & Reversibility

A **quantum circuit** is a sequence of quantum gates arranged over time acting on a set of qubits [cite: 1.1.1, 1.3.3]. Computation flows left-to-right along qubit wires until a measurement operator collapses the state vector into classical output bits [cite: 1.1.1, 1.3.3, 19].

Property	Classical Logic Gates	Quantum Gates
Input / Output States	Discrete bits (0 or 1) [cite: 1.1.1]	Qubit vectors in superposition [cite: 1.1.1, 19]
Reversibility	Mostly Irreversible (AND/OR lose data) [cite: 1.1.1, 1.3.6]	Strictly Reversible (Unitary operations) [cite: 1.1.1, 1.3.4, 1.3.6]
Multi-Qubit Effects	Independent bit interactions	Generates Quantum Entanglement [cite: 1.1.1, 1.3.5, 1.3.6]